Do Now · retrieval

Insertion sort: trace and implement

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is [5,2,4] after one complete ascending bubble-sort pass?

Q2

What is the value of i after i++ when i was 2?

Q3

Is a stack FIFO or LIFO?

Q4

What change makes bubble sort terminate early on sorted data?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

One key at a time — The prefix to the left of i is already sorted; take a[i] as the key.

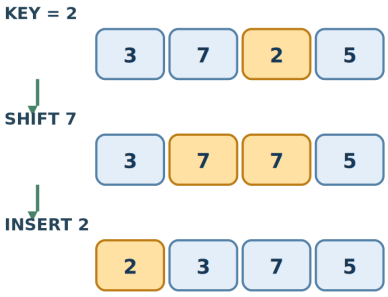
Core idea 2

Make room — Shift larger values in the sorted prefix one position to the right.

Core idea 3

Insert the key — Place it at j+1; the sorted prefix grows by one element.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

key — Value being inserted into the sorted prefix.

sorted prefix — Initial segment already placed in order.

shift — Move a value one index to create space.

in-place — Uses little additional data storage.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

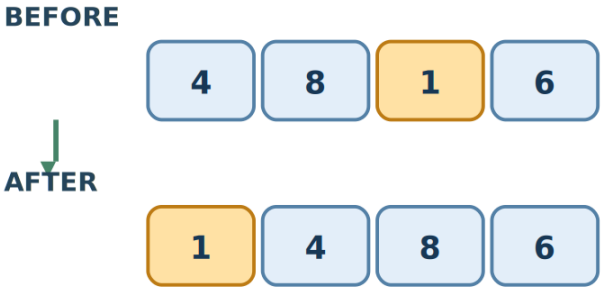
Problem

Insert key 1 into the sorted prefix [4,8] of [4,8,1,6].

Follow the worked method

1. Predict first — Identify the shifts needed.
2. Model solution — Shift 8 then 4 to the right and insert 1 at index 0 → [1,4,8,6].

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Which value must move first?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Insert key 3 into [2,6,3,8], assuming [2,6] is sorted.

Question 2

Trace insertion sort fully on [5,1,4].

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why insertion sort is efficient on almost-sorted input.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Insertion sort always swaps each element with its neighbour.

Use key vocabulary from memory

1. Explain key in your own words.

2. Explain sorted prefix in your own words.

3. Explain shift in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.